

Syllabus

Matter Around Us, Rocks and Minerals, Food and Health, Diseases, Plants Around Us, Animals Around Us, Human Body, Work and Energy, Force and Simple Machines, Sun, Moon and Earth, Air and Water, Pollution and Natural Disasters.

Instructions:

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. Plants are amazing because they make their own food! They use sunlight, water from the soil, and a gas from the air. What is this process called, and what gas do they take in?
(A) Respiration; they take in Oxygen.
(B) Photosynthesis; they take in Carbon Dioxide.
(C) Evaporation; they take in Nitrogen.
(D) Digestion; they take in Sunlight.
2. A hand fan is kept still on a table. When a student starts moving the fan back and forth, air is felt on the face and the body feels cooler. Which statement best explains what is happening?
(A) The fan creates energy from nothing
(B) The fan converts muscular energy into mechanical energy that moves air
(C) The fan changes heat energy into sound energy
(D) The fan stores electrical energy inside it
3. Why do animals like frogs and ducks have "webbed feet"?
(A) To help them climb trees better.
(B) To help them push against the water and swim more effectively.
(C) To help them run faster on dry land.
(D) To keep their feet warm in cold mud.

4. Imagine you are eating a delicious apple. As you chew, your teeth break the food into smaller pieces and mix it with saliva. Which part of your body does the food go to next, and what happens there?
- (A) It goes to the heart to be turned into blood.
- (B) It goes through the food pipe (esophagus) into the stomach where it is mixed with digestive juices.
- (C) It goes straight to the lungs to give us energy from the air.
- (D) It goes to the brain to help us think better.
5. Rohan loves eating only burgers, fries, and chocolate every single day. He refuses to eat fruits or vegetables. After a few weeks, his teacher notices that his gums are bleeding when he brushes his teeth. Which vitamin is Rohan likely missing?
- (A) Vitamin A, which helps him see in the dark.
- (B) Vitamin D, which makes his bones strong.
- (C) Vitamin C, which keeps gums healthy and helps wounds heal.
- (D) Vitamin K, which helps stop bleeding after a cut.
6. Our blood travels through different types of "pipes" in our body. One type of pipe (Arteries) carries blood away from the heart, while another type (Veins) brings blood back to the heart. Why do veins have special "valves" inside them that arteries do not have?
- (A) To make the blood flow faster.
- (B) To prevent the blood from flowing backward, especially when moving up against gravity.
- (C) To filter out germs from the blood before it reaches the heart.
- (D) To give the blood its red color.
7. Why is wind considered a renewable source of energy?
- (A) Wind is produced only at night
- (B) Wind can be created by machines
- (C) Wind is naturally replenished and does not get exhausted
- (D) Wind contains stored electrical energy
8. **Assertion (A):** If a person has a fracture, they should not try to move the injured part.
Reason (R): A fracture is a break or a crack in a bone, and moving it can cause more damage to the bone and surrounding muscles.
- (A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

- (B) Both Assertion and Reason are true, but the Reason is NOT the correct explanation of the Assertion.
- (C) Assertion is true, but the Reason is false.
- (D) Assertion is false, but the Reason is true.
9. A hydroelectric power plant uses flowing water to produce electricity. Which combination of energy changes is involved?
- (A) Heat $\rightarrow$ Light $\rightarrow$ Sound (B) Electrical $\rightarrow$ Mechanical $\rightarrow$ Heat
- (C) Mechanical $\rightarrow$ Electrical (D) Chemical $\rightarrow$ Heat $\rightarrow$ Light
10. In a crowded city with heavy traffic, scientists notice a rise in average local temperatures and a frequent grey haze that irritates people's eyes. Which combination of pollutants is most likely responsible?
- (A) Oxygen and nitrogen (B) Water vapor and argon
- (C) Carbon dioxide and smog (D) Helium and neon
11. **Assertion (A):** People living in very cold regions, like the Arctic, eat a lot of fat-rich food.
- Reason (R):** Fats provide more than double the energy of carbohydrates and help create a layer of insulation under the skin to keep the body warm.
- (A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
- (B) Both Assertion and Reason are true, but the Reason is NOT the correct explanation of the Assertion.
- (C) Assertion is true, but the Reason is false.
- (D) Assertion is false, but the Reason is true.
12. A student tries to open a tight jar lid using bare hands but fails. Then the student uses a cloth and opens it easily. Which type of force mainly helped in opening the lid?



- (A) Gravitational force (B) Magnetic force
- (C) Frictional force (D) Muscular force only

- 13.** Farmers often use chemicals to protect their crops from insects. However, when these chemicals are used excessively, they remain in the ground. Over time, these toxins can be absorbed by the roots of plants and eventually reach humans who eat the vegetables.

Based on the passage, what is a long-term risk of excessive chemical use in farming?

- (A) It makes the soil more fertile for the next year.
 - (B) It causes beneficial insects like earthworms to grow faster.
 - (C) Harmful chemicals can enter the human food chain.
 - (D) It prevents the water cycle from occurring.
- 14.** You are exploring a jungle and find the skeleton of an animal. You observe that it has long, sharp, curved canine teeth and its eye sockets face forward. What can you conclude about the animal?
- (A) It was a herbivore that ate plants and fruits.
 - (B) It was a carnivore that hunted other animals for food.
 - (C) It lived in water and fed on aquatic plants.
 - (D) It was a bird that lost its beak and developed teeth.
- 15.** Which statement best explains why a pulley fixed to a well is called a simple machine?
- (A) It increases the speed of work
 - (B) It changes the direction of applied force
 - (C) It reduces the weight of the load
 - (D) It stores energy
- 16.** A 200 ml sample of a substance is transferred from a cubic container into an open 500 ml spherical bowl. The substance settles to the bottom, taking the shape of the bowl's base, but its volume remains 200 ml. This substance is _____
- (A) Solid
 - (B) Liquid
 - (C) Gas
 - (D) Plasma
- 17.** A screwdriver is used to tighten a screw into a wooden plank. Which combination of simple machines is involved?



- (A) Lever and pulley
- (B) Wheel and axle with screw
- (C) Inclined plane and wedge
- (D) Pulley and wedge

18. Anjali wants to start a vegetable garden. She needs soil that can hold enough water for the plants but also allows some air to reach the roots. Which type of soil should she choose?

- (A) Gravelly soil (B) Clayey soil (C) Loamy soil (D) Rocky soil

19. Match the observations in the table below to determine which row correctly identifies the type of change occurring:

Scenario	Observation	Type of Change
I	An ice cube turns into a puddle	Chemical
II	A piece of wood is burnt to ash	Chemical
III	Salt is dissolved in a glass of water	Chemical
IV	A silver spoon is polished	Chemical

Which scenario is correctly labeled?

- (A) I (B) II (C) III (D) IV

20. Imagine you are looking at two different animals: a **Cow** and a **Tiger**. Which of the following statements about their digestive systems and teeth is correct?

- (A) The Tiger has long, flat teeth to grind grass and a very long stomach.
 (B) The Cow has sharp, pointed teeth (canines) to tear meat and a short stomach.
 (C) The Cow has broad, flat teeth to grind plants and a complex stomach to digest tough grass.
 (D) Both animals have the exact same type of teeth because they are both mammals.

21. Why is the Sun considered the most important member of the solar system?

- (A) It is the smallest object
 (B) It moves around the Earth
 (C) It provides heat and light energy to all planets
 (D) It reflects light from the Moon

22. When a spoonful of salt is stirred into a beaker of water until it completely disappears, the resulting saltwater is called ____.

- (A) Solute (B) Solvent (C) Solution (D) Sediment

- 23. Assertion:** Planting more trees helps in reducing the amount of carbon dioxide in the atmosphere.
Reason: During the process of photosynthesis, trees absorb carbon dioxide to prepare their food and release oxygen.
- (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
- (C) Assertion is true, but Reason is false.
- (D) Assertion is false, but Reason is true.
- 24.** Which of the following is NOT a healthy way to keep germs away from our bodies?
- (A) Trimming our nails regularly so dirt doesn't get trapped inside.
- (B) Taking a bath every day to keep our skin clean.
- (C) Coughing into the open air so the germs can fly away.
- (D) Washing our hands after using the toilet.
- 25.** You are about to eat some fruits and vegetables from the market. You notice some dust and tiny dirt on them. Why should you wash them carefully before eating?
- (A) To make them look clean and colorful
- (B) To remove germs and dirt that can make us sick
- (C) To make them taste sweet
- (D) To make them grow bigger
- 26.** The Earth takes 24 hours to complete one rotation on its axis. How many rotations does the Earth complete in 3 days?
- (A) 24 (B) 3 (C) 48 (D) 72
- 27.** A group of students planned to place a small solar cooker in their school playground. To get maximum heat for cooking food, at which time of the day should they use the solar cooker?
- (A) Early morning, when the Sun is rising
- (B) Late evening, when the Sun is setting
- (C) Afternoon, when the Sun is almost overhead
- (D) Night time, when the Moon is bright
- 28.** On a sunny afternoon, a puddle on the sidewalk completely disappears. Which process of the water cycle is primarily responsible for this?
- (A) Condensation (B) Precipitation (C) Evaporation (D) Infiltration

- 29.** During a full Moon night, no solar eclipse can occur. This is because:
- (A) The Earth is between the Sun and the Moon
 - (B) The Moon is between the Sun and the Earth
 - (C) The Sun is not visible at night
 - (D) The Moon does not reflect sunlight
- 30.** If you were to trap a sample of clean, dry air in a jar, which gas would occupy approximately 78% of the space inside?
- (A) Oxygen
 - (B) Carbon dioxide
 - (C) Argon
 - (D) Nitrogen
- 31.** During a camping trip, a group of students finds a clear-looking stream. To ensure the water is safe from invisible biological impurities like bacteria before drinking, the most reliable immediate step is ____.
- (A) Decantation
 - (B) Boiling the water for 10 minutes
 - (C) Passing it through a cloth filter
 - (D) Letting it sit for an hour
- 32.** Meera found a bird's nest and observed the bird's beak. The beak was very strong, short, and shaped like a cone. Based on the shape of the beak, what kind of food does this bird most likely eat?
- (A) It uses its beak to sip nectar from long, deep flowers.
 - (B) It uses its beak to catch and tear the flesh of smaller animals.
 - (C) It uses its beak to crack open hard seeds and grains.
 - (D) It uses its beak to scoop up fish from the water.
- 33.** Mosquitoes are more than just annoying; they can be dangerous because they are "Vectors." What does this mean?
- (A) They are very fast fliers.
 - (B) They carry germs (like Malaria or Dengue) from a sick person to a healthy person.
 - (C) They eat other harmful insects.
 - (D) They only live in clean, dry places.
- 34.** Which of the following methods would be most effective for a student trying to separate a mixture of iron nails and small glass beads?
- (A) Sieving
 - (B) Magnetic Separation
 - (C) Handpicking
 - (D) Filtration

(A) An igneous rock (B) A sedimentary rock

(C) A metamorphic rock (D) A volcanic rock

1) B	2) B	3) B	4) B	5) C	6) B	7) C
8) A	9) C	10) C	11) A	12) C	13) C	14) B
15) B	16) B	17) B	18) C	19) B	20) C	21) C
22) C	23) A	24) C	25) B	26) B	27) C	28) C
29) A	30) D	31) B	32) C	33) B	34) B	35) A